R V College of Engineering
Department of Computer Science and Engineering(data science and cyber security)
ADLDCO CIE - I : Question Paper

Course:(Code) CS234AI	Applied digital logic design and computer organisation		Semester : 3rd BE
Date : Oct 2024	Duration : 120 minutes	Staff :DD/PSB/KS/KBR/KB	
Name :	USN :	Section :	A/B/C/D/E/CD/CY

PART A

Sl. No	Quiz questions	Marks	* L1-L6	*CO
1.1	What are the minimum number of 2 input NAND gates required to implement the function A ⊕B ⊕C	1	L2	CO2
1.2	Draw the truth table for a system which accepts 4 inputs and gives a high output when more than two inputs are high.	1	L2	CO2
1.3	Subtract the following numbers using 2's complement. 15 from 13	1	L2	CO2
1.4	Construct the Booth recording of a multiplier for the following sequence. 1 0 1 0 1 1 1 1 1 1 1 1 1 0 1 0 1 0 1 1 0	1	L2	CO2
1.4	Simplify the following expression and write the prime implicates and essential prime implicates. F(a,b,c,d)= πM(0,1,4,5,7,8,9,11,14,15)_dc(2,3)	2	L2	CO2
1.5	With example, specify the significance of the overflow in binary addition and binary subtraction?	2	L2	CO1
1.6	Represent the number + 35.25 using 32 bit(single precision) floating point representation.	2	L2	CO1

PART B

Sl. No	Answer all the questions	Marks	* L1-L6	*CO
1a	A temperature control system has four inputs: A (Temperature High), B (Temperature Low), C (Humidity High), and D (System Active). The system should activate under these conditions: When the temperature is high and humidity is low (A=1,B=0,C=0). When the system is active regardless of temperature and humidity (D=1). When the temperature is low and humidity is high (B=1,C=1). Create a truth table, fill in the K-map and find the simplified expression for the system activation condition.	4	L3	CO2
b	Design a single circuit for BCD to excess 3 and excess.3 to BCD converter using 4 bit parallel adder and appropriate gates)	6	L3	CO2
2a	Using Modified Booth's multiplication algorithm, multiply +9 with -6. (Note: +9 is the multiplicand and (-6) is the multiplier)	4	L3	CO1
b	Illustrate the design of a single decade BCD adder(decimal adder) with correction network and show the truth table	6	L2	CO2
3	Explain the process for calculating carry in a Carry Look Ahead adder, design a four bit carry look ahead adder(derive the equations) How does	10	L3	CO2

PTO

	propagation delay in a Carry Look ahead Adder compare to that of a parallel adder?			
4	A home security system has four sensors: A (Garage Door Open), B (Front Door Open), C (Back Door Open), and D (Alarm Active). The alarm activates in the following cases: If any door is open (at least one of A,B, or C is 1). If the alarm is already active (D=1). **Task:** Develop a truth table, and using Quine Mc-cluskey method, obtain the essential prime implicants for generating the alarm.	10	L4	CO2
5	Draw the flow chart for restoring division algorithm and perform the division operation on the following unsigned integers:Dividend:11 Divisor:5	10	L3	CO1

Course Outcomes: After completing the course, the students will be able to:-	
CO 1	Apply design requirements for digital systems and Computer organization
CO 2	Analyze the models used for designing various Combinational and Sequential circuits
CO 3	Develop applications of synchronous sequential networks using flip flops, registers and counters
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CO 5	Investigate techniques of digital system design for building industry relevant real-world systems using electronic components and modern tools

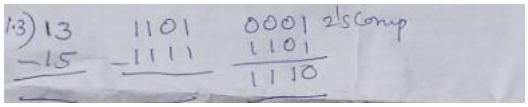
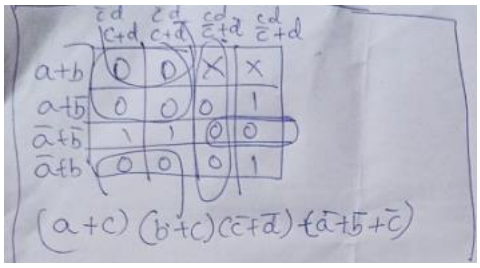
Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	18	42					16	34	10		

	R V College of Engineering Department of Computer Science and Engineering(data science and cyber security) ADLDCO CIE - I : Question Paper and scheme and solution	
Course: (Code)CS234 AI	Applied digital logic design and computer organisation	Semester : 3rd BE

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PART A

Sl. No	Quiz questions	Marks	* L1-L6	*CO
1.1	What are the minimum number of 2 input NAND gates required to implement the function A ⊕ B ⊕ C 8 Nand gates	1	L2	CO2
1.2	Draw the truth table for a system which accepts 4 inputs and gives a high output when more than two inputs are high. □ The output Y is set to **1** when there are more than two inputs that are high (1). □ Specifically, the combinations that yield a high output are: • A=0, B=1, C=1, D=1 (3 high) • A=1, B=0, C=1, D=1 (3 high) • A=1, B=1, C=0, D=1 (3 high) • A=1, B=1, C=1, D=0 (3 high) • A=1, B=1, C=1, D=1 (4 high) **A B C D Y (Output)** 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 1 1 0 0 1 0 0 0 0 1 0 1 0 0 1 1 0 0 0 1 1 1 1	1	L2	CO2

	1 0 0 0 0 1 0 0 1 0 1 0 1 0 0 1 0 1 1 1 1 1 0 0 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 1			
1.3	Subtract the following numbers using 2's complement. 15 from 13 	1	L2	CO2
1.4	Construct the Booth recording of a multiplier for the following sequence. 1 0 1 0 1 1 1 1 1 1 1 1 1 0 1 0 1 0 1 1 0 Soln: -1 +1 -1 +1 0 0 0 0 0 0 0 0 0 -1 +1 -1 +1 -1 +1 0 -1 0	1	L2	CO2
1.5	Simplify the following expression and write the prime implicates and essential prime implicates. F(a,b,c,d)= πM(0,1,4,5,7,8,9,11,14,15)_dc(2,3)  Ploting on kmap-1m and EPI-1m	2	L2	CO2
1.6	With example, specify the significance of the overflow in binary addition and binary subtraction? Overflow occurs in binary addition when the result of adding two binary numbers exceeds the range that can be represented with a fixed number of bits. **Example**: consider an 8-bit binary system: • **Binary Numbers**: ○ A = `01111111` (127 in decimal)	2	L2	CO1

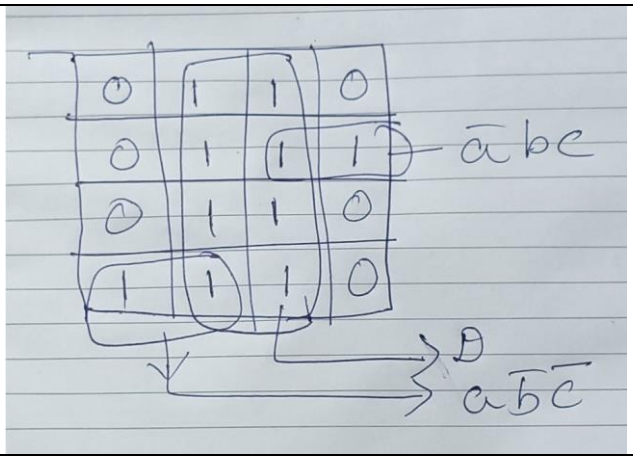
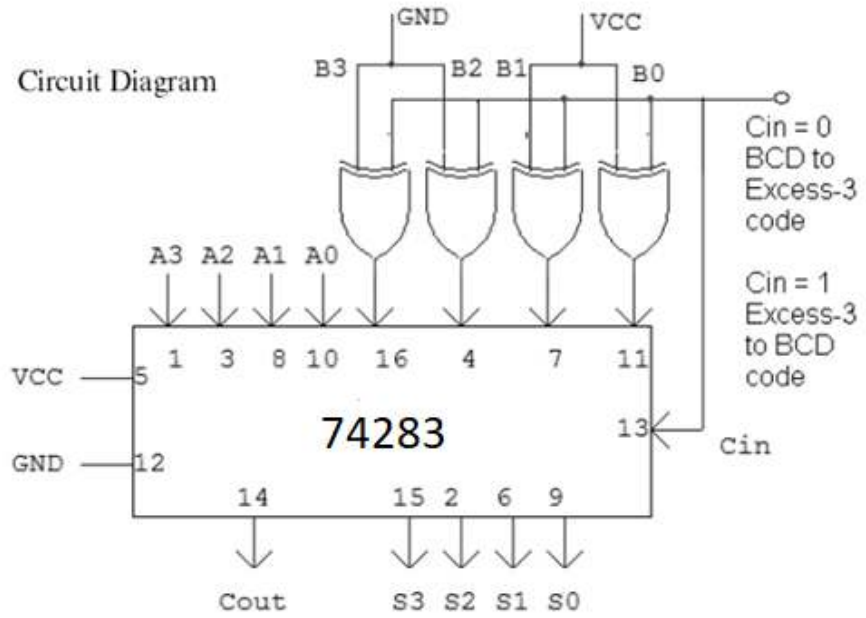
	○ B = `00000001` (1 in decimal)
	
	Addition:
	
	`  01111111`
	`+ 00000001`
	`----------`
	`  10000000  =1mark`
	
	Overflow in binary subtraction occurs when the result of subtracting two numbers yields a result that is outside the representable range.
	
	Example: Consider the same 8-bit system:
	
	• **Binary Numbers**:
	○ A = `00000000` (0 in decimal)
	○ B = `00000001` (1 in decimal)
	
	A = 00000000 (0 in decimal) B = 01111111 (127 in decimal)
	00000000
	- 01111111
	---------- 	
	10000001 (which is -127 in decimal)
	If A (0) is less than B (127) and the result appears to be a large positive number (`10000001`), it indicates that an underflow occurred because the expected result (negative value) cannot be represented correctly in the range. `=1mark`			

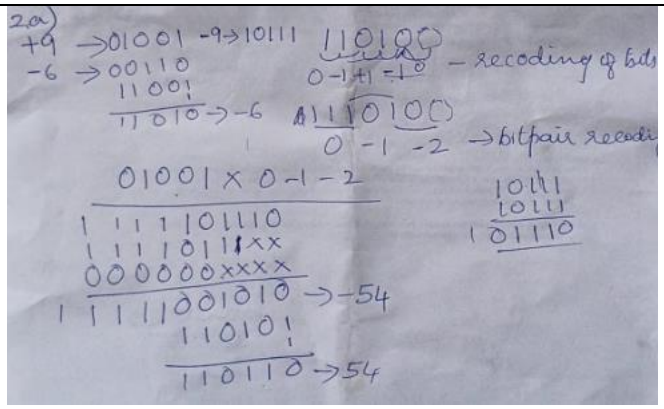
1.7	Represent the number + 35.25 using 32 bit(single precision) floating point representation. 35 in decimal = 100011 in binary. **Convert the fractional part (0.25) to binary**: 0.25 in decimal = 0.01 in binary 35.25 in binary is $100011.01_{(2)}$ We need to express 100011.01100011.01100011.01 in normalized form: Move the binary point 5 places to the left: ○ 1.0001101×2^5 bias is 127, so 127+5=132 ○ 132 in decimal = 10000100100000100100000100 Mantissa: 00011010000000000000000 0 10000100 00011010000000000000000	2	L2	CO1

PART B

Sl. No	Answer all the questions	Marks	* L1-L6	*CO
1a	A temperature control system has four inputs: A (Temperature High), B (Temperature Low), C (Humidity High), and D (System Active). The system should activate under these conditions: When the temperature is high and humidity is low (A=1,B=0,C=0). When the system is active regardless of temperature and humidity (D=1). When the temperature is low and humidity is high (B=1,C=1). Create a truth table, fill in a K-map,	4	L3	CO2

	and find the simplified expression for the system activation condition. Conditions for Activation 1. **Condition 1**: A = 1, B = 0, C = 0 (Temperature High, Humidity Low) 2. **Condition 2**: D = 1 (System Active) 3. **Condition 3**: B = 1, C = 1 (Temperature Low, Humidity High) **A B C D Y (System Active)** 0 0 0 0 0 0 0 0 1 1 0 0 1 0 0 0 0 1 1 1 0 1 0 0 0 0 1 0 1 1 0 1 1 0 1 0 1 1 1 1 1 0 0 0 1 1 0 0 1 1 1 0 1 0 0 1 0 1 1 1 1 1 0 0 0 1 1 0 1 1 1 1 1 0 0 1 1 1 1 1 Y=D+AB′C′+A’BC			

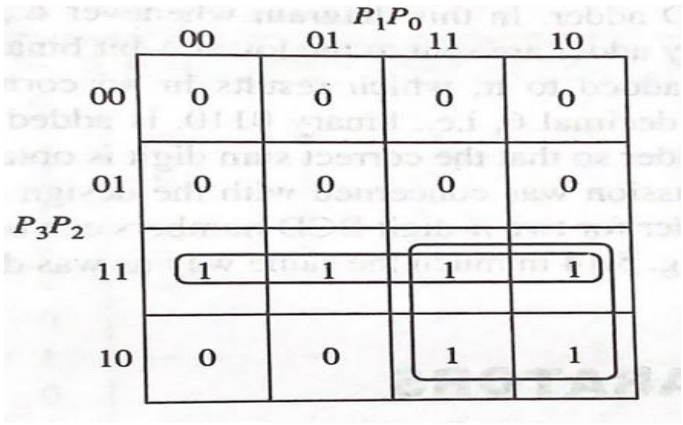
				
b	Design a single circuit for BCD to excess 3 and excess 3 to BCD converter using 4 bit parallel adder and appropriate gates) **BCD → EXCESS-3** 0 0 0 0 | 0 0 1 1 0 0 0 1 | 0 1 0 0 0 0 1 0 | 0 1 0 1 0 0 1 1 | 0 1 1 0 0 1 0 0 | 0 1 1 1 0 1 0 1 | 1 0 0 0 0 1 1 0 | 1 0 0 1 0 1 1 1 | 1 0 1 0 1 0 0 0 | 1 0 1 1 1 0 0 1 | 1 1 0 0 **EXCESS-3 → BCD** 0 0 1 1 | 0 0 0 0 0 1 0 0 | 0 0 0 1 0 1 0 1 | 0 0 1 0 0 1 1 0 | 0 0 1 1 0 1 1 1 | 0 1 0 0 1 0 0 0 | 0 1 0 1 1 0 0 1 | 0 1 1 0 1 0 1 0 | 0 1 1 1 1 0 1 1 | 1 0 0 0 1 1 0 0 | 1 0 0 1 Truth table -2 marks, circuit with brief explanation-4 marks 	6	L3	CO2
2a	Using Modified Booth's multiplication algorithm, multiply +9 with -6. (Note: +9 is the multiplicand and (-6) is the multiplier)	4	L3	CO1

				
b	Illustrate the design of a single decade BCD adder(decimal adder) and show the truth table	6	L2	CO2

Comparting binary and BCDsums

Decimal Sum	Binary sum					Required BCD sum				
	K	P_3	P_2	P_1	P_0	C_{out}	Z_3	Z_2	Z_1	Z_0
0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	1	0	0	0	0	1
2	0	0	0	1	0	0	0	0	1	0
3	0	0	0	1	1	0	0	0	1	1
4	0	0	1	0	0	0	0	1	0	0
5	0	0	1	0	1	0	0	1	0	1
6	0	0	1	1	0	0	0	1	1	0
7	0	0	1	1	1	0	0	1	1	1
8	0	1	0	0	0	0	1	0	0	0
9	0	1	0	0	1	0	1	0	0	1
10	0	1	0	1	0	1	0	0	0	0
11	0	1	0	1	1	1	0	0	0	1
12	0	1	1	0	0	1	0	0	1	0
13	0	1	1	0	1	1	0	0	1	1
14	0	1	1	1	0	1	0	1	0	0
15	0	1	1	1	1	1	0	1	0	1
16	1	0	0	0	0	1	0	1	1	0
17	1	0	0	0	1	1	0	1	1	1
18	1	0	0	1	0	1	1	0	0	0
19	1	0	0	1	1	1	1	0	0	1

=2m

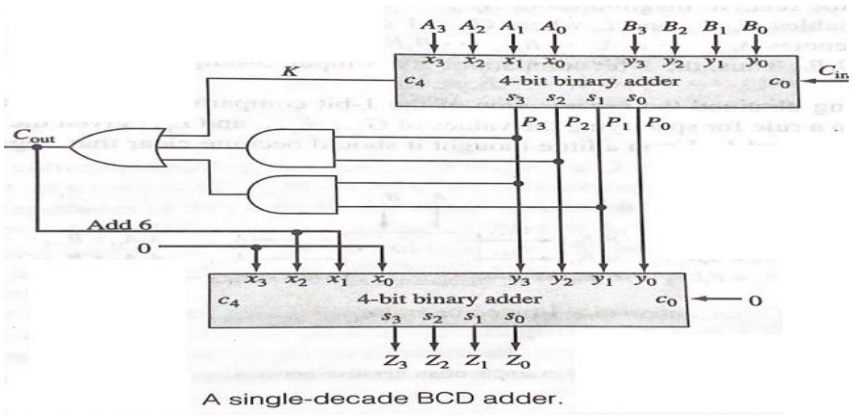
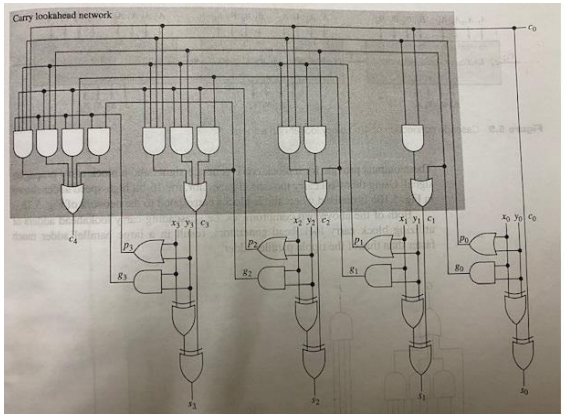


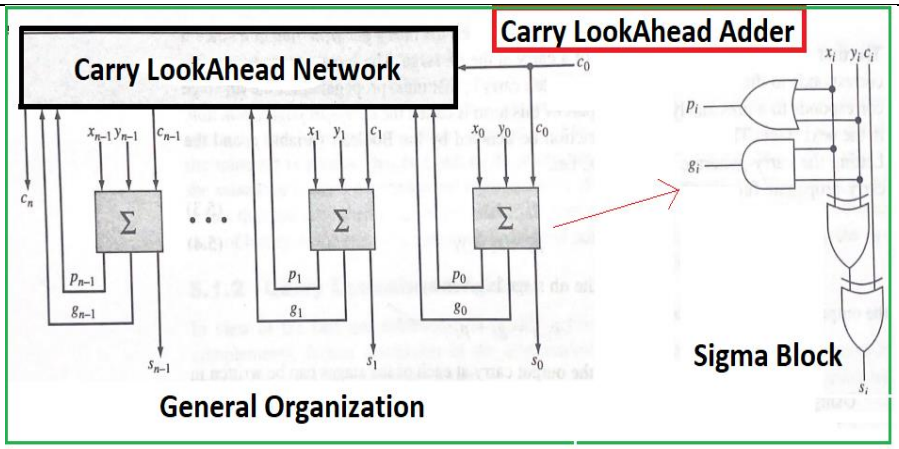
KMap to detect the combinations

1010,1011,1100,1101,1110,1111

from the Kmap

ADD_6 = P3P2 + P3P1

	Including the Carry output (K), in the expression, the Boolean expression for adding the correction is ADD_6 = K + P3P2 + P3P1 =2m  A single-decade BCD adder. =2m			
3	Explain the process for calculating carry in a Carry Look Ahead adder, design a four bit carry look ahead adder(derive the equations) How does propagation delay in a Carry Look ahead Adder compare to that of a parallel adder? **Si = Xi .xor. Yi .xor. Ci** **Ci+1 = Xi Yi + Xi Ci + Yi Ci** **= Xi Yi + Ci (Xi + Yi)** **= Xi Yi + Ci (Xi .or. Yi)** **Let us represent, XiYi by Gi (also called as Carry generate function) & Xi.or.Yi by Pi (also called carry propagate function)** **= Gi + Ci Pi** **C1 = G0 + P0 C0** **C2 = G1 + P1 C1 = G1 + P1 G0 + P1 P0 C0** **C3 = G2 + P2 C2 = G2 + P2 G1 + P2 P1 G0 + P2 P1 P0 C0** **C4 = G3 + P3 C3 = G3 + P3 G2 + P3 P2 G1 + P3 P2 P1 G0 + P3 P2 P1 P0 C0 = 5 marks**  carry look ahead adder for 4 bit opernads is shown in the figure. Generalizing from the figure, The path length from the generation of carry to its apperance as an input at any higher order stage, i.e the path length through any stage of the carry look ahead network, is two levels of logic. Thus with one level of logic to from Gi, two levels of logic for the carry to propogate between any two stages, and one level of logic to have the carry effect a sum output, the maximum propogation delay for CLA is 4 Units of time, assuming each gate introduces a unit of time of propogation delay. <- 4Bit CLA Desgin Circuit-3marks Propagation delay comparision-2 marks Or	10	L3	CO2

				
4	A home security system has four sensors: A (Garage Door Open), B (Front Door Open), C (Back Door Open), and D (Alarm Active). The alarm activates in the following cases: If any door is open (at least one of A,B, or C is 1). or If the alarm is already active (D=1). Consider for both if conditions also **Task**: Develop a truth table, and using Quine Mc-cluskey method, obtain the essential prime implicants for generating the alarm. To create a truth table for the home security system based on the revised conditions, we need to clarify how the alarm activates under the specified conditions: Inputs: - A: Garage Door Open (1 if open, 0 if closed) - B: Front Door Open (1 if open, 0 if closed) - C: Back Door Open (1 if open, 0 if closed) - D: Alarm Active (1 if active, 0 if inactive) Conditions for Alarm Activation: The alarm activates if: 1. At least one door is open: (A or B or C) (true if any of A, B, or C is 1) 2. Or if the alarm is already active: (D = 1)Final Logical Expression: Thus, the new alarm state (D= (A or B or C) or D This means that the alarm will be active if: - Any door (A, B, or C) is open - The alarm is already active	10	L4	CO2

Truth Table:

A	B	C	D	Dactive
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

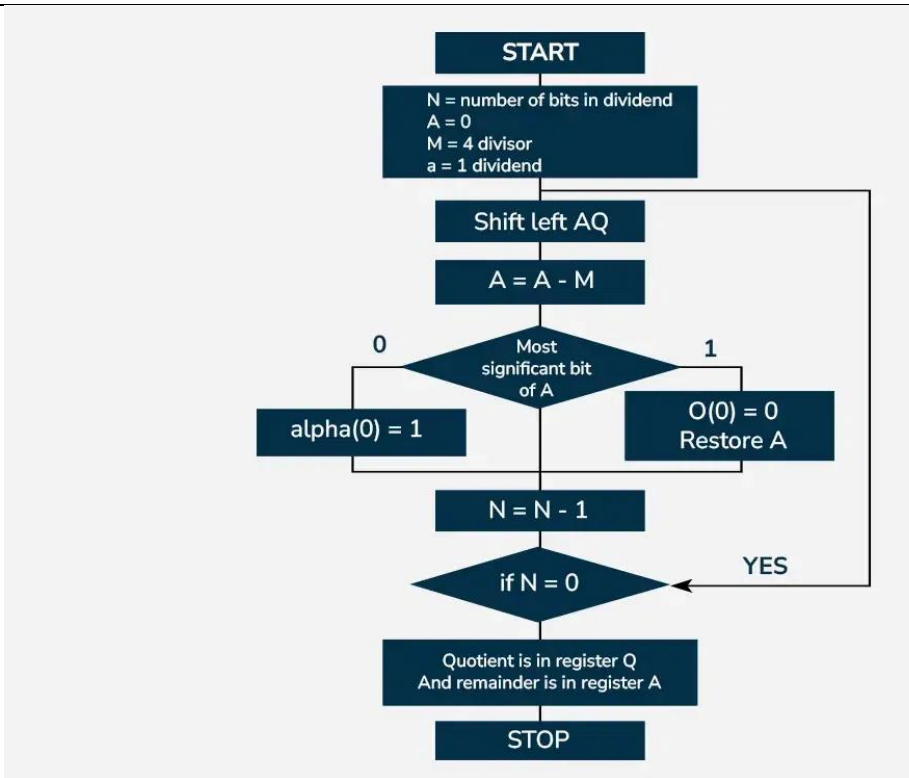
System activates when A+B+C+D is active

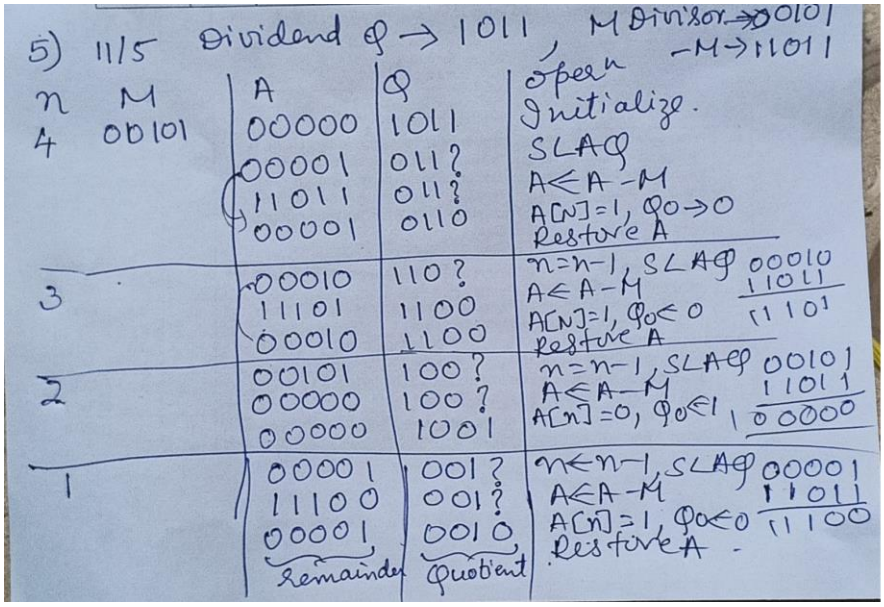
For And logic=m(3,5,7,9,11,13,15) System activates when A+B+C+D is active

ad+bd+cd

| A | B | C | D | Dactive |

	|---|---|---|---|----------| | 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 0 | 1 | 1 | | 0 | 0 | 1 | 0 | 1 | | 0 | 0 | 1 | 1 | 1 | | 0 | 1 | 0 | 0 | 1 | | 0 | 1 | 0 | 1 | 1 | | 0 | 1 | 1 | 0 | 1 | | 0 | 1 | 1 | 1 | 1 | | 1 | 0 | 0 | 0 | 1 | | 1 | 0 | 0 | 1 | 1 | | 1 | 0 | 1 | 0 | 1 | | 1 | 0 | 1 | 1 | 1 | | 1 | 1 | 0 | 0 | 1 | | 1 | 1 | 0 | 1 | 1 | | 1 | 1 | 1 | 0 | 1 | | 1 | 1 | 1 | 1 | 1 |	0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1		
5	Draw the flow chart for restoring division algorithm and perform the division operation on the following unsigned integers:Dividend:11 Divisor:5	10	L3	CO1





Flow chart 4+ solving 6 marks

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Marks Distribution	Test	Max Marks	18	42					16	34	10		

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PART A

Sl. No	Quiz questions	Marks	* L1-L6	*CO
1.1	Develop a logic circuit to implement the functionality of an XOR gate utilizing a 2:1 multiplexer, demonstrating the integration of combinational logic with minimal hardware resources.	1	L3	CO2
1.2	Calculate the maximum achievable operating frequency of a MOD-32 ripple counter, given that the propagation delay per flip-flop is 50 ns, while considering the cumulative effect of sequential flip-flop delays on the circuit's performance.	1	L3	CO3
1.3	What techniques can be employed to mitigate the issue of "0" and "1" catching in digital circuits, ensuring reliable signal integrity and robust operation?	1	L3	CO3
1.4	Derive the Characteristic Equation for JK Flip-Flop	1	L2	CO3
1.5	A D-Flip Flop can be converted into T-Flip Flop by using additional logic circuit**,** Obtain the Boolean expression for additional logic circuit	1	L3	CO3
1.6	Why are asynchronous inputs referred to as overriding inputs, and how do they directly affect the state of a flip-flop irrespective of clock signals?	1	L3	CO3
1.7	Draw the circuit diagram of a 4:16 Decoder using 3:8 decoder	2	L2	CO3
1.8	Draw the Logic diagram for Master-Slave Flip-Flop with Data Lockout and How can a data lock-out flip-flop be symbolically distinguished from a pulse-triggered flip-flop in digital circuit representations?	2	L2	CO3

PART B

Sl. No	Answer all the questions	Marks	* L1-L6	*CO
1a	Design and implement an asynchronous Mod-6 up counter, detailing the circuit configuration, flip-flop connections, and state transitions, and provide the corresponding clock timing diagram that illustrates the sequential counting behavior	5	L3	CO3
b	Elucidate the operational principles of a synchronous 2N twisted ring counter, detailing the sequential state	5	L3	CO3

	transitions, feedback mechanism, and the role of flip-flops in generating the desired counting sequence			
2a	Design a digital logic circuit to implement the Boolean functions F1(a,b,c)=πM(0,3,5,6,7) and F2 (a,b,c)=πM(2,3,4,5,7) using a 3-to-8 decoder, ensuring minimal usage of input gates. Provide a step-by-step analysis of how the decoder outputs can be utilized to construct the functions efficiently, while optimizing the number of gates used for combining the relevant terms.	5	L3	CO2
b	Design and implement the Boolean function f(A,B,C,D)=∑m(0,1,2,4,6,9,12,14) using a 4-to-1 multiplexer. Provide a detailed explanation of how to map the min terms to the MUX inputs, and outline the necessary logic for selecting the appropriate input lines based on the function's truth table.	5	L3	CO2
3a	Design a 4bit Universal shift register for the operations specified in the table S1 S0 Register Operation 0 0 No Change 0 1 Shift Right 1 0 Shit Left 1 1 Load the parallel data	5	L2	CO3
b	Detail the working of 1:4 Demultiplexer and with this, Design and implement a 1:8 demultiplexer utilizing two 1:4 demultiplexers, ensuring correct signal routing through hierarchical selection lines. Provide a detailed circuit configuration that demonstrates how to cascade the two 1:4 demultiplexers, managing the selection logic to achieve the required output channels.	5	L3	CO2
4a	Explain the phenomenon of a race-around condition in sequential circuits, particularly in JK flip-flops, and describe how this issue is mitigated using a master-slave JK flip-flop configuration. Additionally, illustrate the use of a gated SR latch to control the input signals, providing a detailed logic diagram and timing waveforms to demonstrate the elimination of the race-around condition in the master-slave configuration.	5	L3	CO3
b	Propose the block diagram to realize a 4-bit Comparator, using 1 bit comparator with iterative approach, with all the connections and indicating clearly the meaning of the inputs and outputs. Demonstrate, the working of the circuit with example data of 4 bit numbers. (No truth table and derivations required)	5	L3	CO3
5a	Design a synchronous counter using positive edge triggered JK flip flops to count 0→1→2→4→5→6→0→1→2....	5	L2	CO3
b	Design and implement an edge-triggered D flip-flop, detailing the circuit configuration, input-output	5	L2	C03

	relationships, and the control mechanisms that ensure synchronization with the clock signal.			

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Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks		16	44				20	40			

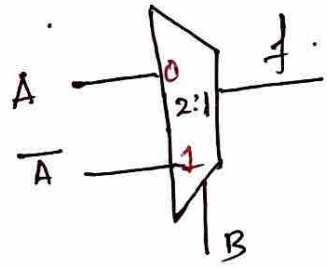
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ADLD & CO

CIE-II Scheme and Solution

PART-A QUIZ

1.1 NOR gate using 2:1 MUX.



—— (1) Marks.

1.2 MOD 32 need 5 flipflop $\because 2^5 = 32$

$$f_{max} = \frac{1}{50ns \times 5} = 4MHz$$ —— (1) Mark.

1.3 0's and 1's catching problem can be resolved with the help of Edge Triggered D-FlipFlop: this FlipFlop consider the Data at the time of Edge and it won't consider the Entire clock level. —— (1) Marks.

1.4 JK FlipFlop Equation

$$Q^+ = J\overline{Q} + \overline{K}Q$$ —— (1) Mark.

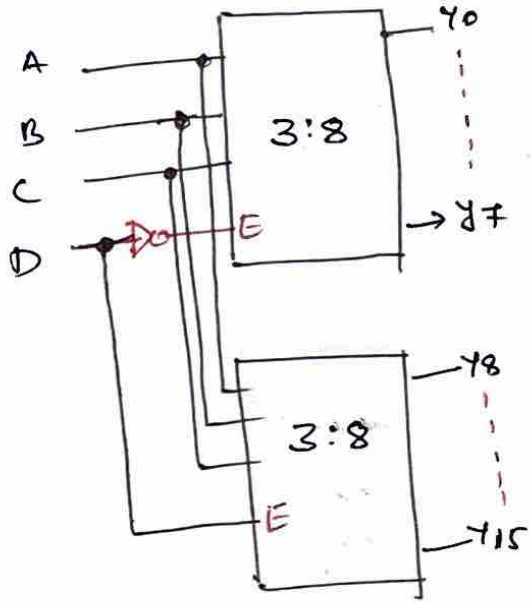
1.5 $D = \overline{T}Q + T\overline{Q}$

$= T \oplus Q$ —— (1) Mark

1.6 Asynchronous inputs are directly connected to a bistable element; hence these inputs are overriding other inputs and also impact the state of a flipflop irrespective of clock signal.

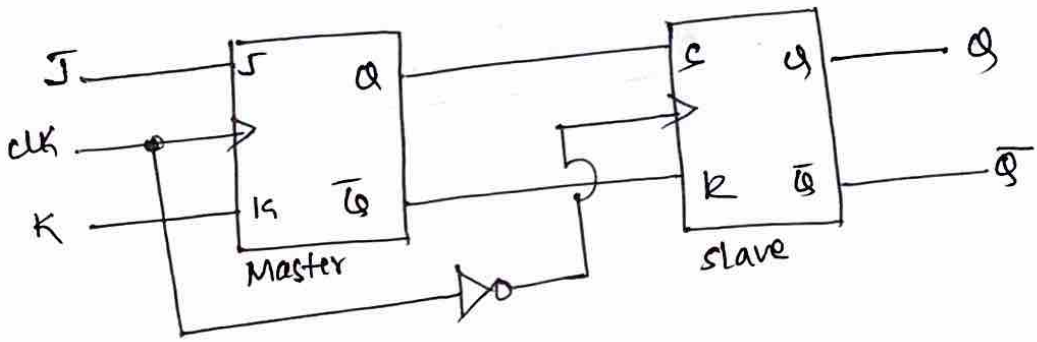
— (1) Mark

1.7

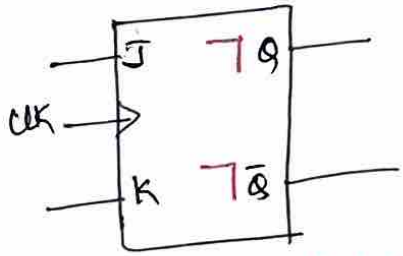


— (2) Marks

1.8



— (2) Marks



Data lockout Representation

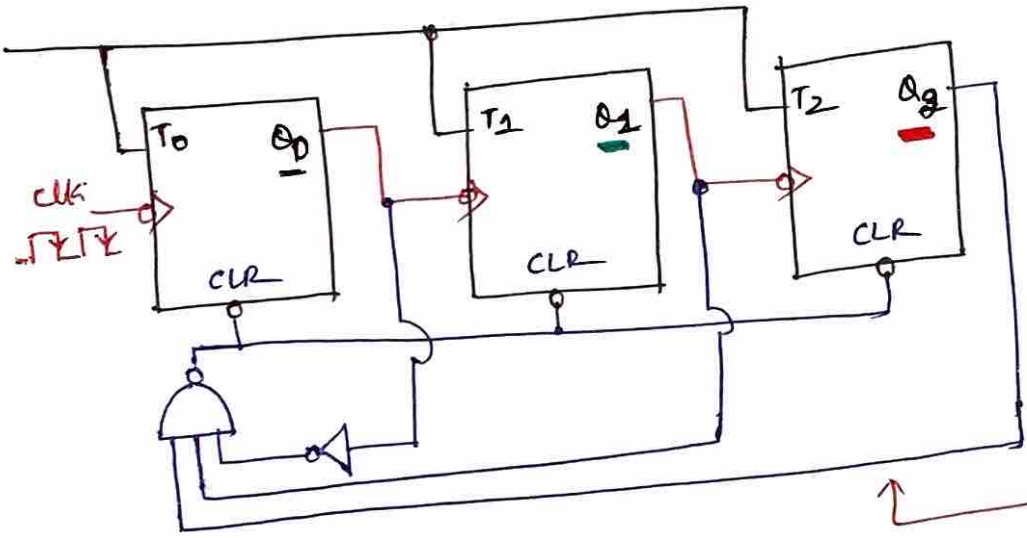
PART-B TEST

1.a MOD6 up counter (Asynchronous counter)

$2^n \geq N$

$2^3 \geq 6$

$8 \geq 6$ ∴ we need 3 Flip Flops



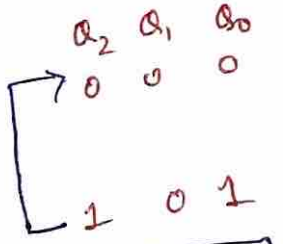
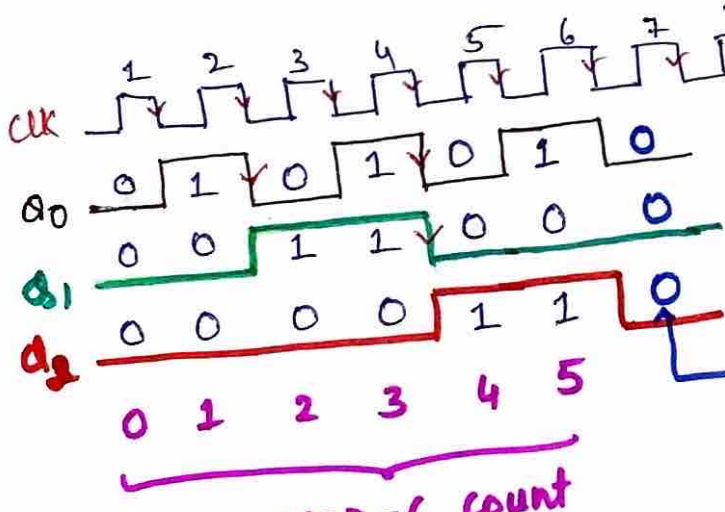


Diagram (3) Marks.



← waveform - (2) Marks

CLR signal takes place.

1.b

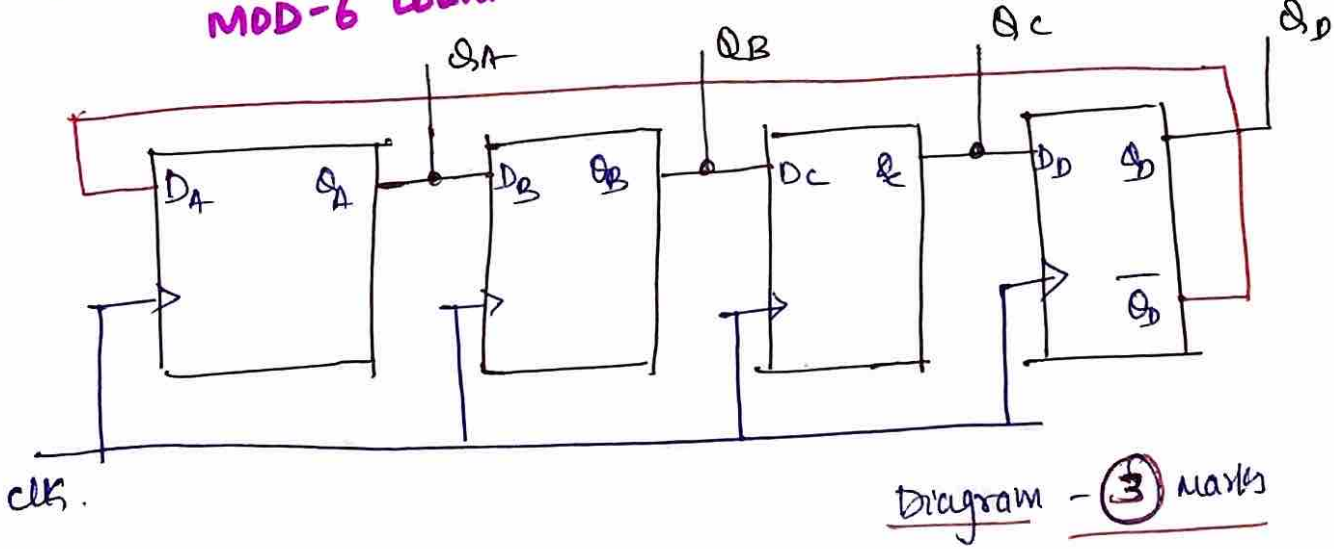


Diagram - (3) Marks

CLK	Q_A	Q_B	Q_C	Q_D	
	0	0	0	0	← Initial state
①	1	0	0	0	
②	1	1	0	0	
③	1	1	1	0	
④	1	1	1	1	
⑤	0	1	1	1	
⑥	0	0	1	1	
⑦	0	0	0	1	
⑧	0	0	0	0	← Initial state

N Bit Johnson Counter

so $N = 4$ ∴

$2N = 2(4) = 8$ counts.

Sequence table — (2) Marks.

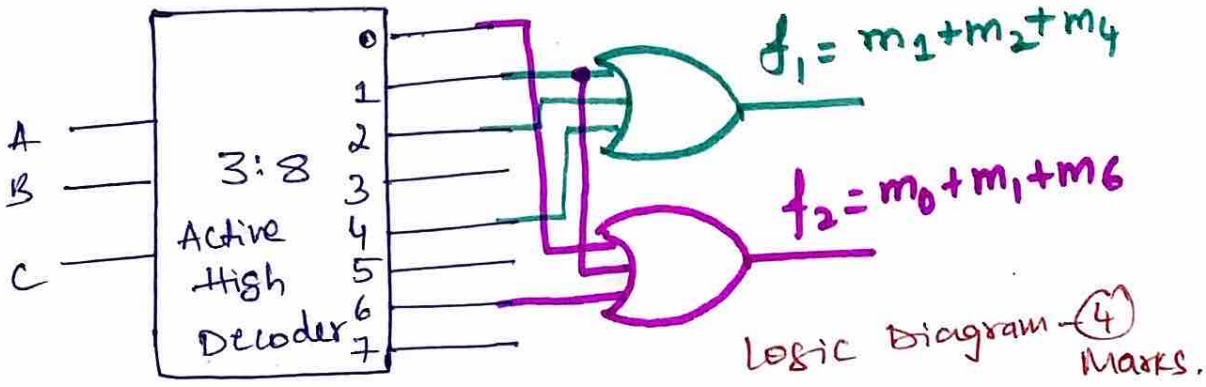
2. a Implement the following function using 3:8 decoder & minimal usage of input gates.

minimal usage of ip gate conversion.

$F_1(a,b,c) = \pi M(0,3,5,6,7) \Rightarrow \Sigma_m(1,2,4)$

$F_2(a,b,c) = \pi M(2,3,4,5,7) \Rightarrow \Sigma_m(0,1,6)$

(1) Mark



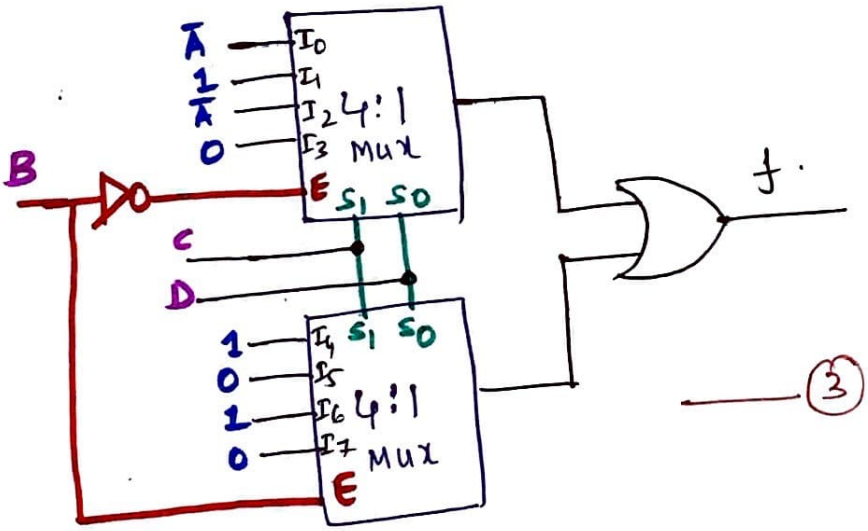
Logic Diagram — (4) Marks.

NOTE: To achieve minimal usage of input gate, First we need to convert pos form to sop form. then Realize the Equation using Active High Decoder & OR gate.

2.b $F(A, B, C, D) = \Sigma m(0, 1, 2, 4, 6, 9, 12, 14)$ using 4:1 mux.

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	$\bar{A}$	1	$\bar{A}$	0	1	0	1	0

(2) Marks



(3) Marks.

A ← for Mapping

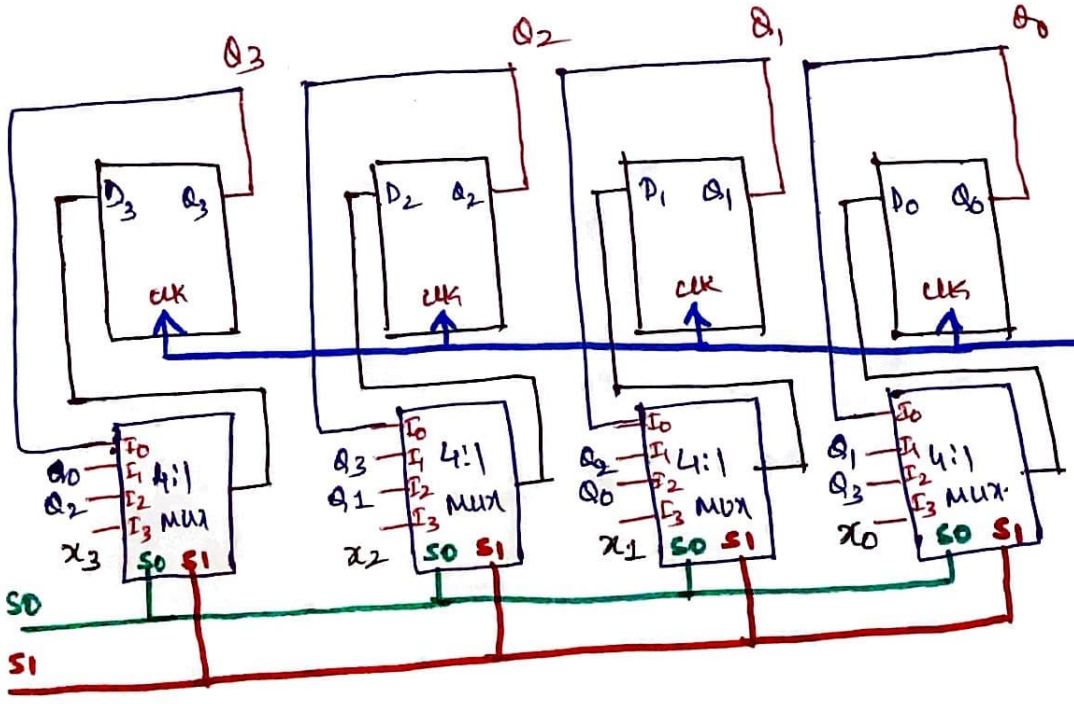
B ← to activate the Enable pin

C & D ~~are~~ ← used for select line.

S_0	S_1	operation
0	0	Hold
0	1	Right shift
1	0	Left shift
1	1	load parallel Data

(1) Mark

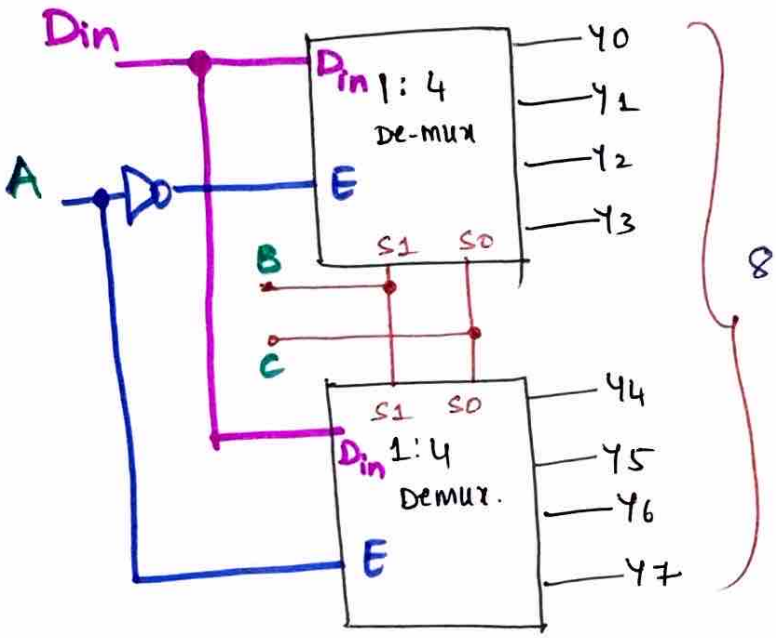
3.a



Logic Diagram

(4) Marks.

3.b



5 Marks.

4.a Race around means continuous toggling. The clock pulse is HIGH for a longer duration than the Flipflop propagation delay, this causes the output to toggle continuously within the same clock, making it Indeterminate

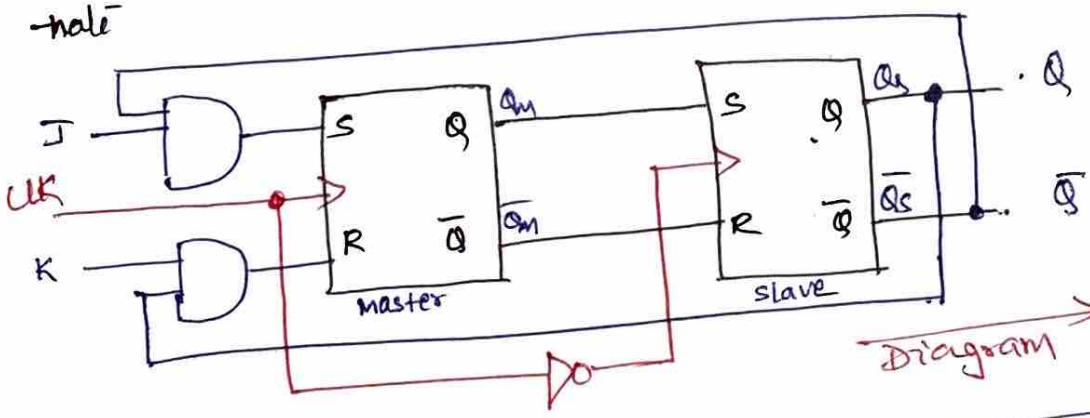
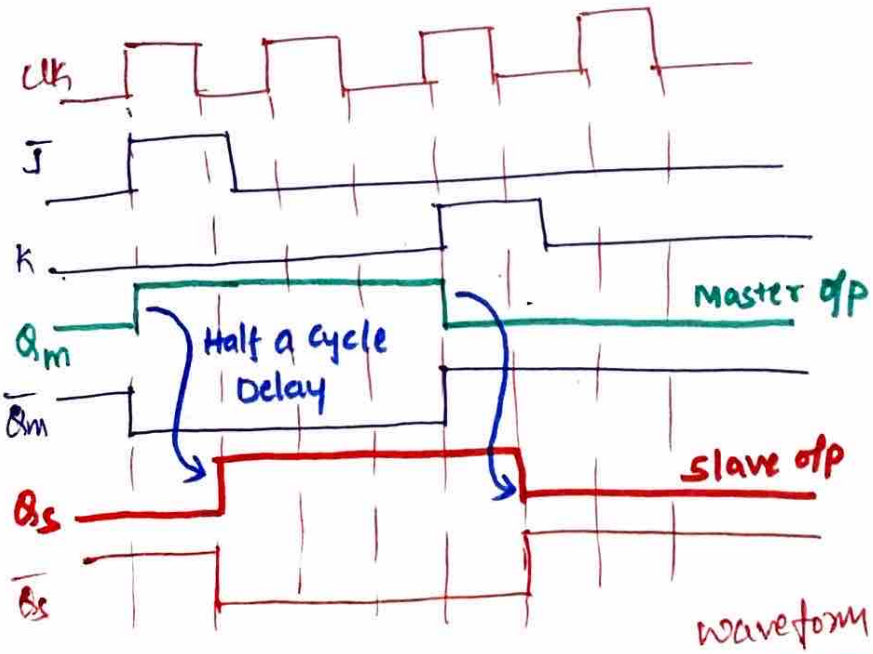


Diagram → 2 Marks

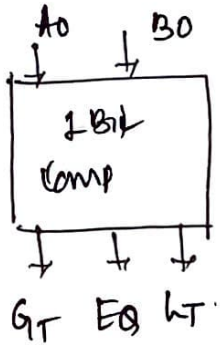
CLK	J	K	Q	$\bar{Q}$	Status
⎍	0	0	Q	$\bar{Q}$	No change
⎍	0	1	0	1	Reset
⎍	1	0	1	0	Set
⎍	1	1	$\bar{Q}$	Q	Toggle
0	x	x	Q	$\bar{Q}$	No change

Truth table → 1 Mark



waveform → 2 Marks

4.b) 1 Bit comparator



A_0	B_0	G_T	E_Q	L_T
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

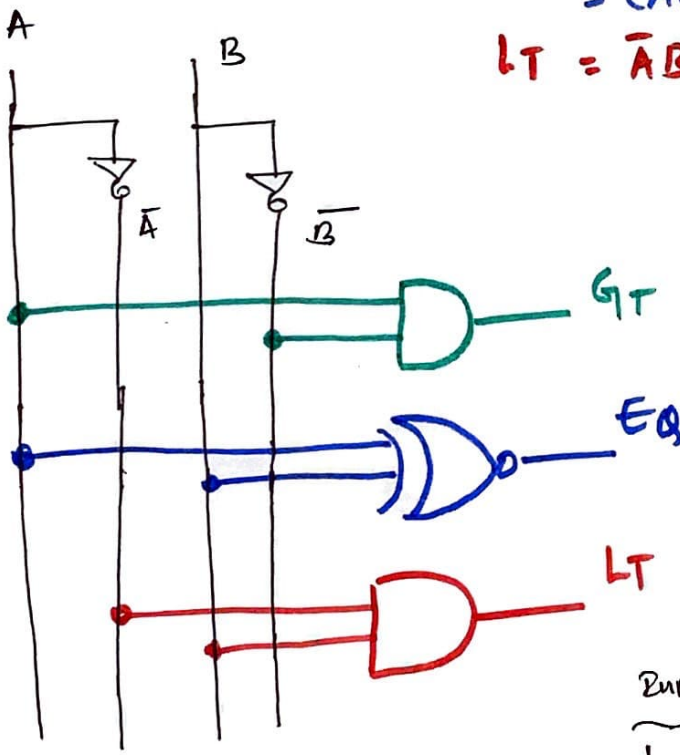
$G_T = A\bar{B}$

$E_Q = \bar{A}\bar{B} + AB$

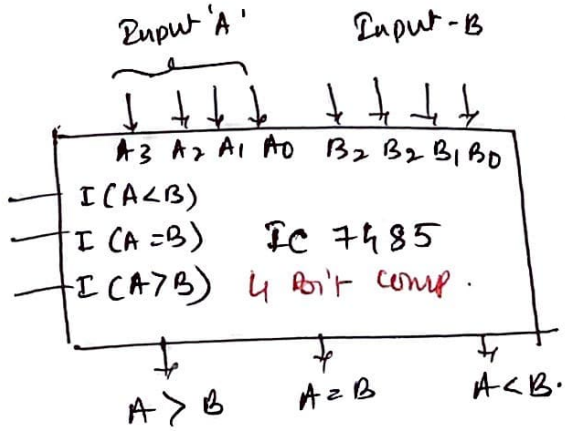
$= \overline{(A \oplus B)}$

$L_T = \bar{A}B$

— 1 Mark



— (2) Marks



— (1) Mark.

4 Bit Example No :— (1) Mark.

total = (5) Marks.

5.a sequence generator.

$0 \to 1 \to 2 \to (3) \to 5 \to (6) \to 0 \to 1 \to 2$

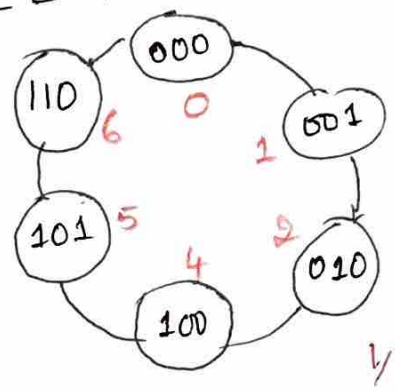
6 → is the highest count ∴ we need (3) Flip flop to construct the above sequence.

JK Excitation table

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

→ 1/2 Marks

State diagram

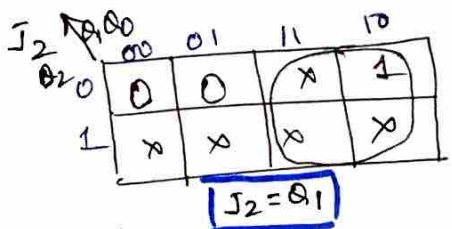


1/2 Mark

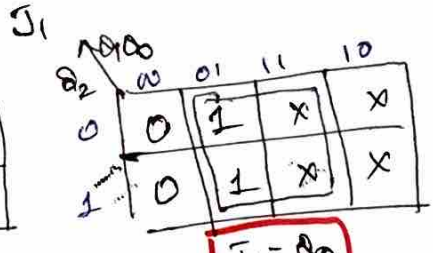
State table

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2 K_2	J_1 K_1	J_0 K_0
0	0	0	0	0	1	0 X	0 X	1 X
0	0	1	0	1	0	0 X	1 X	X 1
0	1	0	0	1	1	0 X	X 0	1 X
0	1	1	1	0	0	1 X	X 1	X 1
1	0	0	x	x	x	X X	X X	X X
1	0	1	1	0	1	X 0	0 X	1 X
1	1	0	1	1	0	X 0	1 X	X 1
1	1	1	0	0	0	X 1	X 1	0 X
			x	x	x	X X	X X	X X

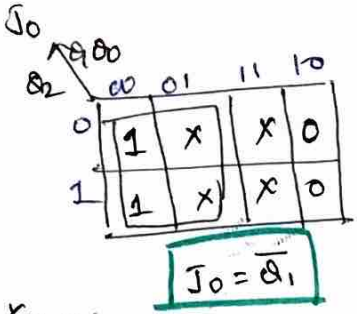
1 1/2 Marks



$J_2 = Q_1$

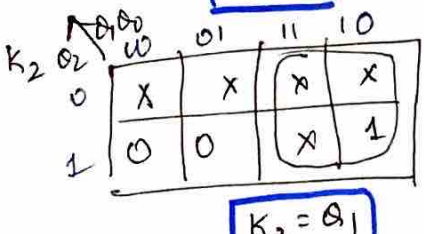


$J_1 = Q_0$

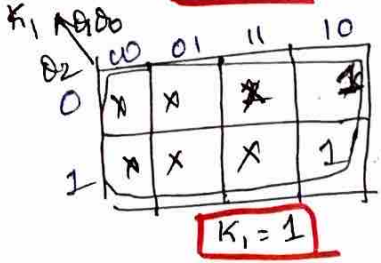


$J_0 = \overline{Q_1}$

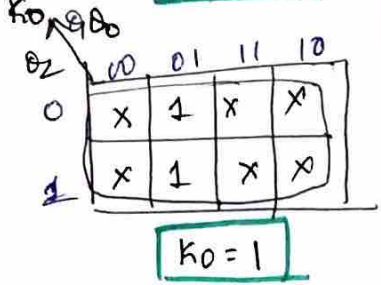
1 1/2 Marks



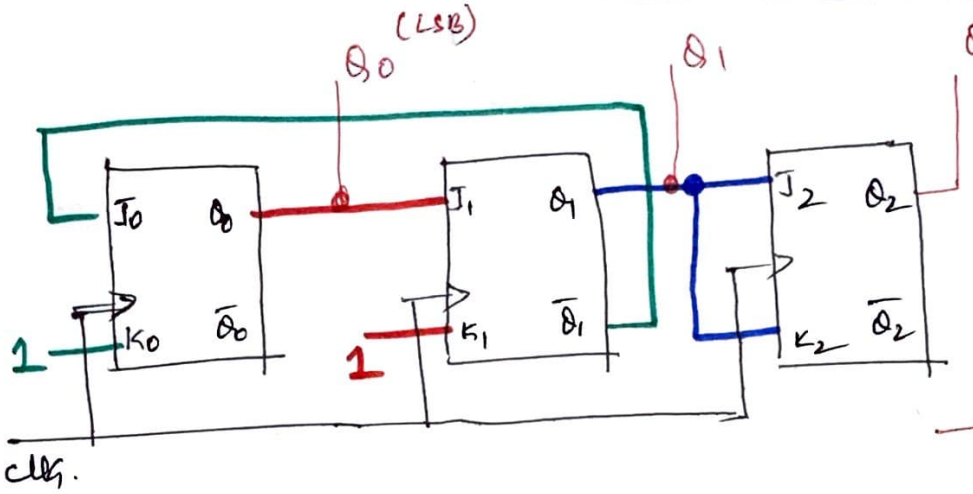
$K_2 = Q_1$



$K_1 = 1$

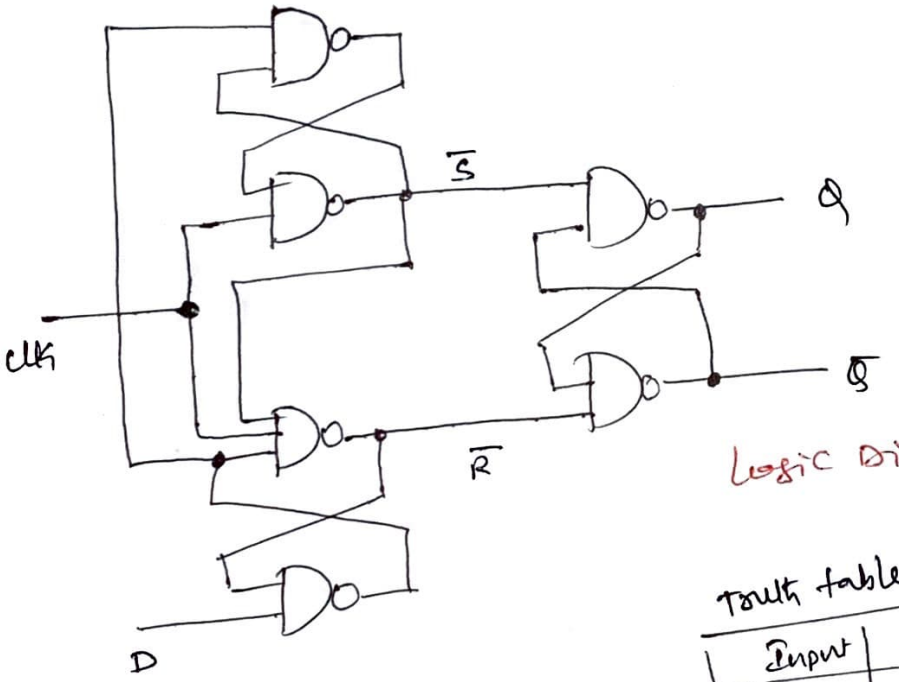


$K_0 = 1$

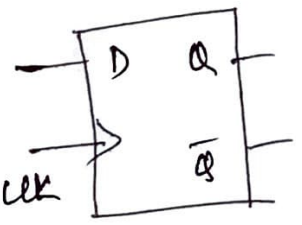


(1) Mark.

5.b



Logic Diagram — (4) Marks



Truth table.

Input		output	
clk	D	Q	$\bar{Q}$
↑	0	0	1
↑	1	1	0
0	x	Q	$\bar{Q}$

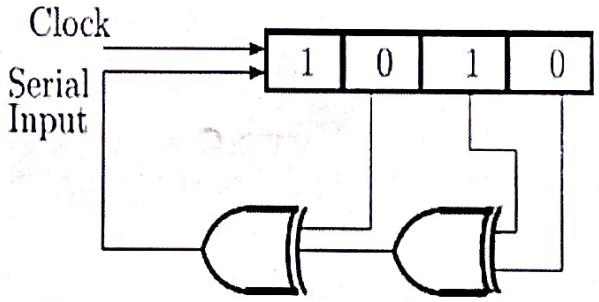
(1) Mark.

	R V College of Engineering Department of Computer Science and Engineering (Data Science & Cyber Security) ADLD & CO Improvement Test & Quz : Question Paper

Course Code CS234AI	**Course Name: Applied Digital Logic Design and Computer Organisation**	Semester : 3rd BE

Date : 7th Jan 2025	Duration : 120 minutes	Staff : DD/PSB/KS/KBR/KB	
Name :	USN :	Section :	A/B/C/D/E/CD/CY

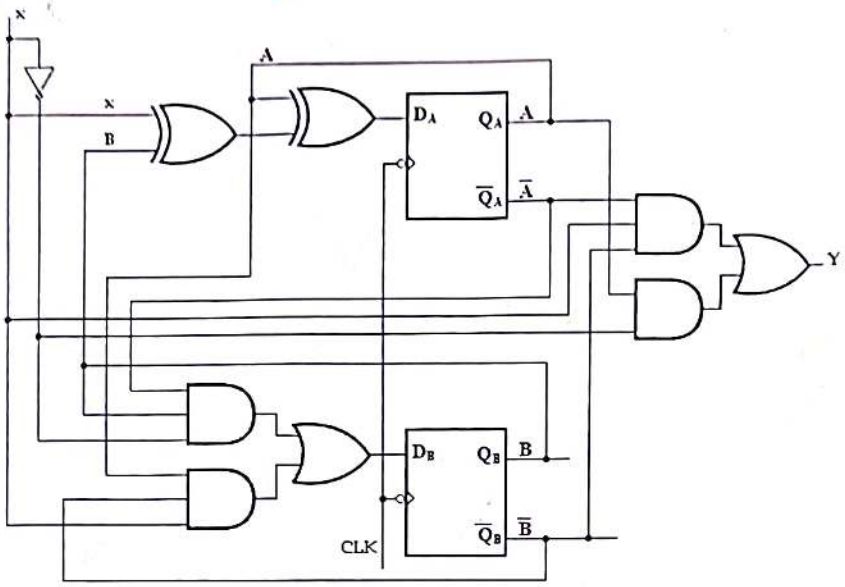
Part-A

Sl. no	Answer all the questions	M	L1-L6	CO
1	Program contains 500 instructions, out of that 40% of instructions requires 1 clock cycles, in the remaining 60%, 50 instructions require 2 clock cycles and the remaining requires 3 clock cycles for execution. Find the total time required to execute the program running on a 1 MHz machine.	2	L2	CO4
2	Draw the block diagram of the Processor/CPU design with Three Bus Datapath.	2	L3	CO4
3	Given program perform the addition of two numbers and indicate the value of the flags CY, Z, OF and N (Note: H means, number is represented in Hex) MOVE #99H, R0 MOVE #8AH, R1 ADD R0, R1,R2 ; R0 + R1 -> R2	2	L3	CO4
4	The shift register shown in the figure is initially loaded with the bit pattern 1010. subsequently the shift register is clocked, and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). The number of clock pulses required to have the content of the shift register 1010 again is___ 	2	L4	CO5
5	Register R1 and R2 of a computer contain the decimal values 3500 and 2800. Find the effective address of the memory operand in the following instructions ADD +(R1), R5 ADD (R1)-, R5	2	L3	CO4

r 35w

3504

Part B

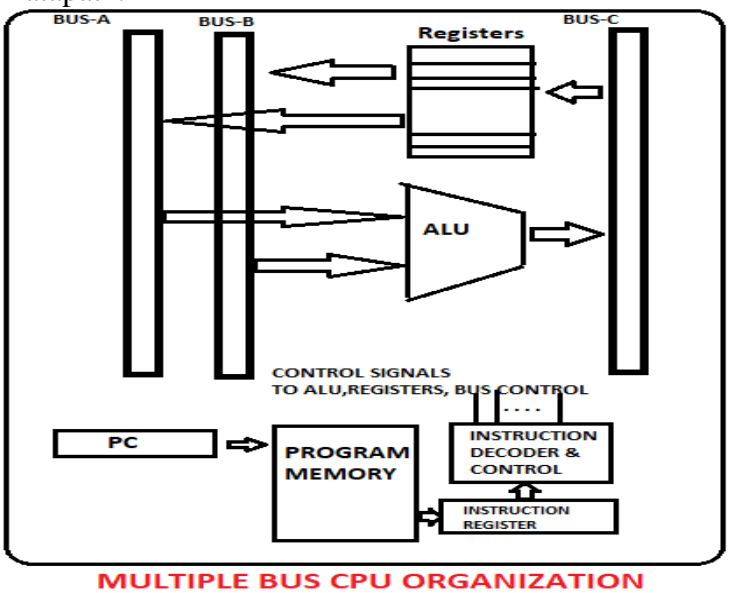
1	Analyze the following circuit and write the excitation equations, derive transition equations, compute excitation table, transition table and draw state diagram. Is this a mealy or a moore model and why?  	10	L4	CO5
2	Design and implement a serial adder using the Moore-type finite state machine (FSM) model, incorporating state transitions, output logic, and input-output synchronization to achieve sequential bit-wise addition of binary numbers.	10	L4	CO5
3	a. Discuss all the addressing Modes supported by the Processor instruction set, with an example instruction. b. Compare processor instructions and assembler directives. Indicate meaning of EQU, DATAWORD, ORG.	6 4	L3	CO4
4	a. Write Assembly Language Code snippets using Three and One address instructions to evaluate the following expression with suitable comments. $Y= X^3 + 24X- 45$ b. Demonstrate with Machine Language / ALP code snippet, using suitable memory layouts (stack frame), the passing and returning of arguments to the subroutine using the stack.	6 4	L4	CO4
5	Write ALP code snippets, for the following scenarios, with clear memory layout (Assume input numbers and the results are stored in the memory, Processor supports direct/indirect operands and two address formats, all the arithmetic and data transfer instructions are supported) a. To find Sum and Average of 'N' 32 bit numbers. b. To find Smallest of two 32 bit numbers	10	L3	CO4

Course Outcomes:

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test/Quiz	Marks				38	22		2	36	22		

	R V College of Engineering Department of Computer Science and Engineering (Data Science & Cyber Security) ADLD & CO Improvement Test & Quz : Question Paper		
Course Code CS234AI	**Course Name: Applied Digital Logic Design and Computer Organisation**		Semester : 3rd BE
Date : 7th Jan 2025	Duration : 120 minutes	Staff : DD/PSB/KS/KBR/KB	
Name :	USN :	Section :	A/B/C/D/E/CD/CY

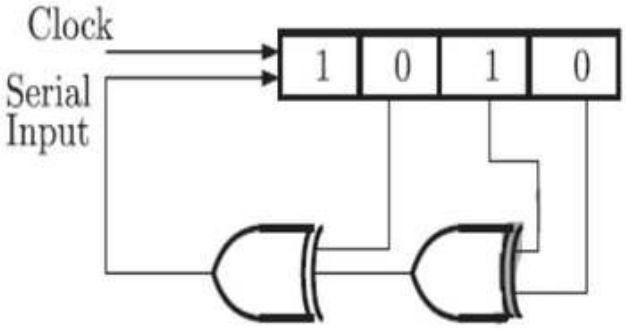
Part-A

Sl. no	Answer all the questions	M	L1-L6	CO
1	Program contains 500 instructions, out of that 40% of instructions requires 1 clock cycles, in the remaining 60%, 50 instructions require 2 clock cycles and the remaining requires 3 clock cycles for execution. Find the total time required to execute the program running on a 1 MHz machine. (40% of 500= 200 x 1 clk cycle = 200 clks Among 60% of 500, i.e 300, = 50 x 2 + 250 x 3 = 850 clks Total clocks = 1050 clks, Time for 1clk = 1/1MHz = 1MicroSecond,) **Answer:** Hence Total Time = 1050 x 1Micro second = 1050 Micro seconds Note: Give Marks, for the answer represented in any units like seconds,nanoseconds,microseconds.	2	L2	CO4
2	Draw the block diagram of the Processor/CPU design with Three Bus Datapath.  MULTIPLE BUS CPU ORGANIZATION Note: Give Marks for Any equivalent diagram / as given in the reference book	2	L3	CO4

<table>
<tr>
<td>3</td>
<td>Given program perform the addition of two numbers and indicate the value of the flags CY, Z, OF and N (Note: H means, number is represented in Hex)
MOVE #99H, R0
MOVE #8AH, R1
ADD R0, R1,R2 ; R0 + R1 -> R2

Assuming 8 bit processor
99H - 1001 1001
8AH - 1000 1010

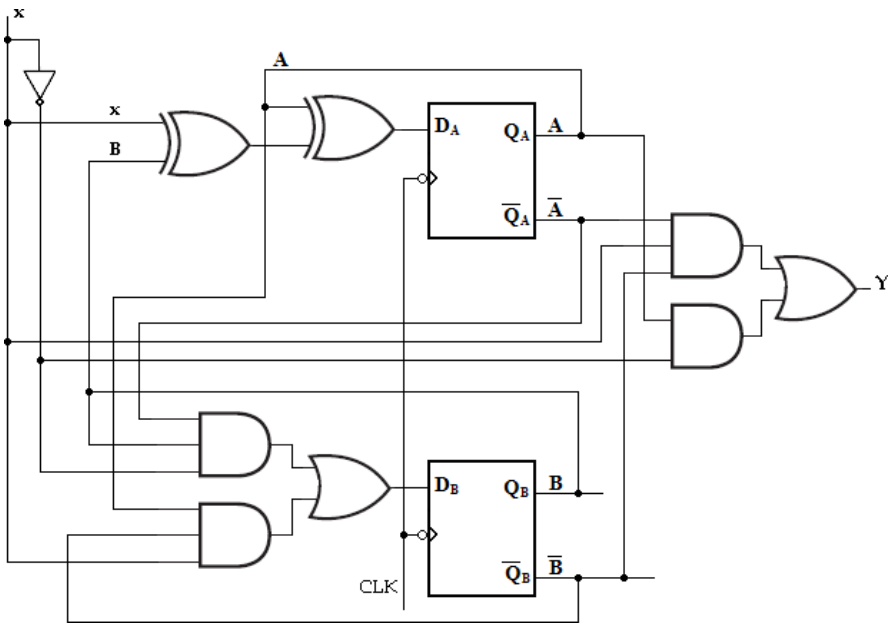
0010 0011

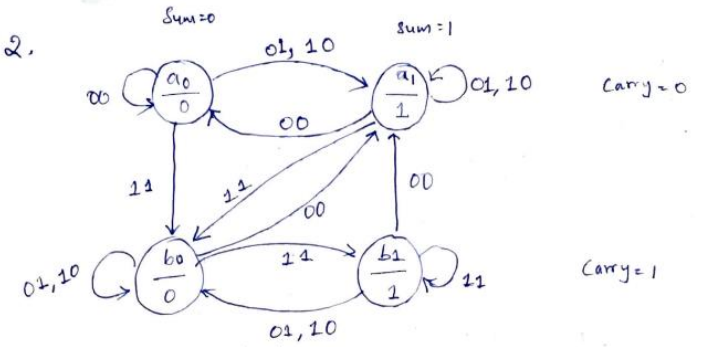
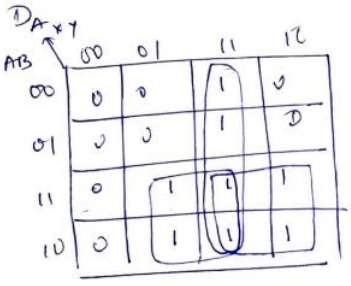
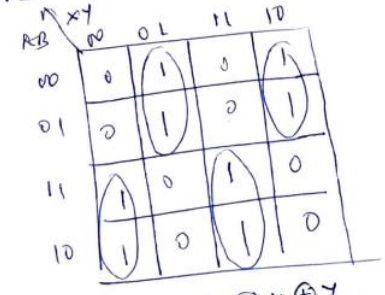
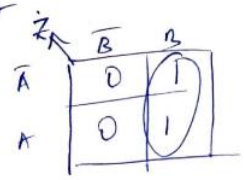
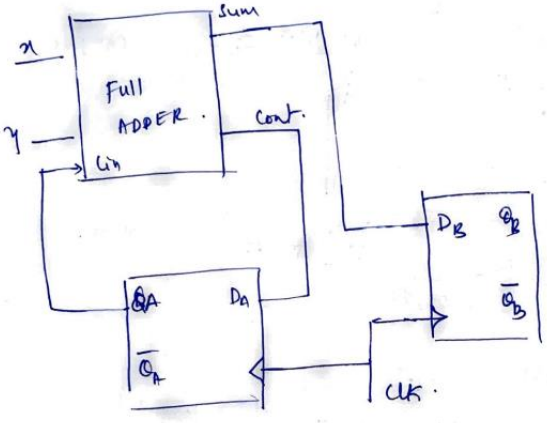
CY=1, Z=0,N=0,OF= Exor(1,0)=1</td>
<td>2</td>
<td>L3</td>
<td>CO4</td>
</tr>
<tr>
<td>4</td>
<td>The shift register shown in the figure is initially loaded with the bit pattern 1010. subsequently the shift register is clocked, and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). The number of clock pulses required to have the content of the shift register 1010 again is___
 

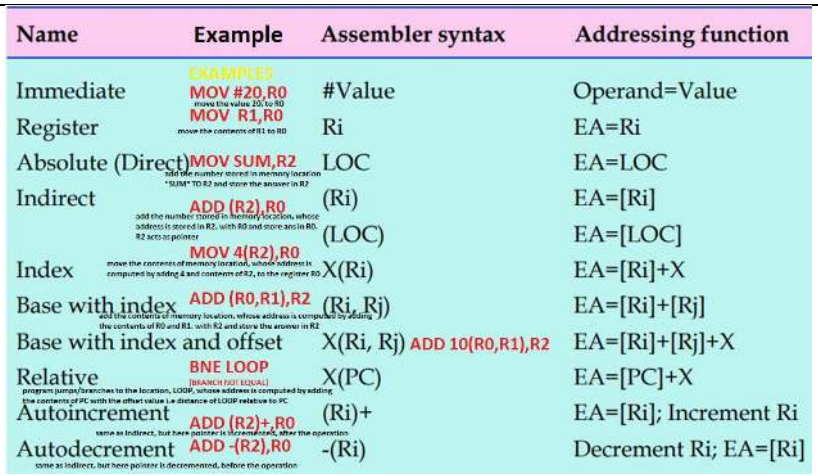
7 Clock Cycles:
Serial Input = Q1.EXOR.(Q2.EXOR.Q3)
Clk Si/p Q0 Q1 Q2 Q3
1 0 1 0
<table>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>4</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>5</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>6</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>7</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr>
</table>
</td>
<td>2</td>
<td>L4</td>
<td>CO5</td>
</tr>
<tr>
<td>5</td>
<td>Register R1 and R2 of a computer contain the decimal values 3500 and 2800. Find the effective address of the memory operand in the following instructions
ADD +(R1), R5
ADD (R1)-, R5
ANSWER TYPE 1: (IF STUDENTS TREAT BOTH INSTRUCTIONS ARE INDEPENDENT AND NOT RELATED) AND IT IS OF 32 BIT CPU/MEM WORD
EA for First Instruction: 3504
(Reason- this addressing mode is of type, pre autoincrement addressing mode, here R1 is a pointer,which increments by 4 and then used for fetching the data.</td>
<td>2</td>
<td>L3</td>
<td>CO4</td>
</tr>
</table>

	Hence, adds used for fetching the data is 3504) **EA for Second Instruction: 3500**. (Reason-this addressing mode is of type,post autodecrement addressing mode, here first data is fetched, and then it will be incremented. So, address used for fetching is 3500.) (IF IT IS 8 BIT CPU/MEMORY WORD – ANS IS 3501/3500. ALWAYS TAKE DEFUALT AS 32 BIT) **ANSWER TYPE 2: (IF STUDENTS TREAT BOTH INSTRUCTIONS ARE RELATED) AND IT IS OF 32 BIT CPU/MEM WORD** EA FOR first instruction: 3504 EA For second instruction:3504 Note: Give Marks for correct anserer, reason is optional.			

Part B

1	Analyze the following circuit and write the excitation equations, derive transition equations, compute excitation table, transition table and draw state diagram. Is this a mealy or a moore model and why?  	10	L4	CO5
2	Design and implement a serial adder using the Moore-type finite state machine (FSM) model, incorporating state transitions, output logic, and input-output synchronization to achieve sequential bit-wise addition of binary numbers.	10	L4	CO5

	2.  State table Present state $Q_A Q_B$ | Next state XY: 00, 01, 10, 11 | output Z a_0 | a_0 a_1 a_1 b_0 | 0 a_1 | a_0 a_1 a_1 b_0 | 1 b_0 | a_1 b_0 b_0 b_1 | 0 b_1 | a_1 b_0 b_0 b_1 | 1  D_A AB | XY: 00 01 11 10 00: 0 0 1 0 01: 0 0 1 0 11: 0 1 1 1 10: 0 1 1 1 $D_A = AX + AY + XY$  D_B AB | XY: 00 01 11 10 00: 0 1 0 1 01: 0 1 0 1 11: 1 0 1 0 10: 1 0 1 0 $D_B = A \oplus X \oplus Y$ Sum.  Z | $\bar{B}$ | B $\bar{A}$ | 0 | 1 A | 0 | 1 $Z = B$.  Logic Diagram.			
3	a. Discuss all the addressing Modes supported by the Processor instruction set, with an example instruction.	6 4	L3	CO4

	 b. Compare processor instructions and assembler directives. Indicate meaning of EQU, DATAWORD, ORG. Answer: Refer Notes In addition to providing a mechanism for representing instructions in a program, assembly language allows the programmer to specify other information needed to translate the source program into the object program. Such statements, called assembler directives (or commands), are used by the assembler while it translates a source program into an object program. Some popular directives are provided below. EQU : used to name the constant values, like #define statements in C N EQU 10 Wherever N is referred in the program, it will replaced by 10, before assembly program is converted to object program ORIGIN: which tells the assembler program, where in the memory to place the instructions that follow. ORG 100 ;It specifies that the instructions of the object program are to be loaded in the memory starting at address 100 DATAWORD: used to declare a variable, which can hold a word, with the initial value N: DATAWORD 150 Value 150 is stored in the variable N, of size one word, and name of the variable is N			
4	a. Write Assembly Language Code snippets using Three and One address instructions to evaluate the following expression with suitable comments. $Y= X^3 + 24X- 45$ 1 address instruction MOVE X MULT X MULT X STORE Z MOVE X	6 4	L4	CO4	

	MULT 24 ADD Z SUB 45 STORE Y 3 adds instruction MULT X,X,Y MULT X,Y,Y MULT 24,X,Z ADD Z,Y,Y SUB 45,Y,Y b. Demonstrate with Machine Language / ALP code snippet, using suitable memory layouts (stack frame), the passing and returning of arguments to the subroutine using the stack. Answer: Part a – any program is ok, only thing is instructions used should satisfy 3adds and 1 adds, and achieve the answer. Not required to be efficient. 3adds: 3M 1adds: 3M Part B: Code snipped, to push one/multiple arguments to stack using Mov/Push, calling a subroutine (pushing return adds on stack),Optional-[save registers,access operand/argument,pop registers,save the result] Return to the main program (pop from stack). Code-2M,Diagram-2M			
5	Write ALP code snippets, for the following scenarios, with clear memory layout (Assume input numbers and the results are stored in the memory, Processor supports direct/indirect operands and two address formats, all the arithmetic and data transfer instructions are supported) a. To find Sum and Average of 'N' 32 bit numbers. b. To find Smallest of two 32 bit numbers Answer: A. Refer Notes/Ref book for adding N numbers, then use DIV instruction / successive subtraction for finding the average (let the student assume the place/reg where quotient is stored). Marks-6Marks B. Use Subtraction/Comparison instruction to compare the two numbers, and use Branch. One such program is given. 4Marks MOVE NUM1,R0 MOVE NUM2,R1 COMPARE R0,R1 ; OR CMP R0,R1 ; RO-R2 is done Branch > 0 GREAT: MOVE R0,RESULT BRANCH SKIP GREAT: MOVE R1,RESULT SKIP: END Note: Give marks for any flag, either Carry or Sign	10	L3	CO4

Course Outcomes:

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test/Quiz	Marks				38	22		2	36	22		

R V College of Engineering

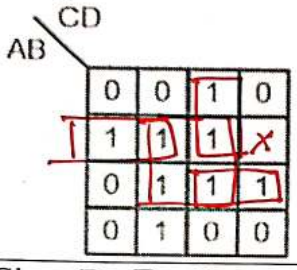
Department of Computer Science and Engineering (data science and cyber security)

Applied Digital Logic Design & Computer Organization

Retest CIE -1 : Question Paper

Course: CS234AI	Applied digital logic design and computer organisation	Semester: 3rd BE	
Date : Dec 2024	Duration : 120 minutes	Staff :DD/PSB/KS/KBR/KB	
Name :	USN :	Section:	A/B/C/D/E/CD/CY

PART A

Sl. No.	Questions	M	BT	CO
1.1	Write down the simplified Boolean expression for the following k-map 	2	3	2
1.2	Given F = ∑m (0, 1, 5, 7, 15, 14, 10), find the number of implicants, Prime implicants and Essential Prime implicants.	2	3	2
1.3	Construct the Booth recording of a multiplier for the following sequence. 0 0 1 1 0 1 0 1 1 0 0 1 1 0 1 0 1 0 1 1 0	2	3	3
1.4	Construct a full adder using 2 half adders.	2	2	4
1.5	What is the range of numbers in 'n' bit 2's complement representation and write the 2's complement representation -28 in 16 bit representation.	2	2	2

PTO

Part B

Sl. No.	Questions	M	BT	CO
1 a)	Perform multiplication of 10101(multiplicand) and 00101 (multiplier)using Booth's algorithm.	5	3	1
b)	Give the flow chart for restoring division and perform the division operation on following number. Dividend: 8 Divisor: 5	5	3	1
2 a)	Design a 4 bit parallel adder using 1bit Full adder and Perform Binary addition on the following numbers. (Assume the numbers are represented in 2's complement form.) i. 23 and -8	5	4	4
b)	Perform the multiplication of 13 X (-8) using Bit pair recording algorithm	5	3	1
3 a)	Consider the decimal number: +105.625. Represent it in equivalent binary representation in IEEE single precision format.	5	3	1
b)	Minimize the following boolean function using Kmap and implement the reduced function using only Nand gates. $F(A, B, C, D) = \Sigma m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$	5	2	2
4	Bring out the design logic behind carry look ahead adder, design a four bit carry look ahead adder and justify the same through algebraic equations.	10	3	1
5	Simplify the expression $F(A,B,C,D) = \sum m(0,1,2,4,6,8,9,11,13,15)$.)using Quine Mc-cluskey method and write the prime implicants and essential prime implicants.	10	3	4

Course Outcomes: After completing the course, the students will be able to:-

CO 1 Apply design requirements for digital systems and Computer organization
CO 2 Analyze the models used for designing various Combinational and Sequential circuits
CO 3 Develop applications of synchronous sequential networks using flip flops, registers and counters
CO 4 Design optimized modern processors and memories for given specifications
CO 5 Investigate techniques of digital system design for building industry relevant real-world systems using electronic components and modern tools

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	11	2	17		9	46	5	-	-

USN | 1 | R | V | 2 | 3 | C | D | 0 | 0 | 3 |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Feb/Mar -2025

Computer Science Engineering

Common to CD / CY / CS

APPLIED DIGITAL LOGIC DESIGN AND COMPUTER ORGANIZATION

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Perform the operation $(+25) + (-18)$ and represent the result in an $8-bit$ signed binary format using two's complement.	02	2	1
	1.2	Multiply -9 and -7 using Booth's Algorithm. Represent the numbers in $5-bit$ binary format.	01	3	2
	1.3	Represent -15.375 in $IEEE$ 754 Single Precision format.	01	2	1
	1.4	In a $2-variable\ K-Map$, how many minterms can be represented?	02	2	1
	1.5	What is the significance of wrapping around the edges in $K-Maps$?	01	1	2
	1.6	How does the Quine-McCluskey method handle don't-care conditions?	02	1	1
	1.7	Show the steps to add two $4-bit$ binary numbers using a binary parallel adder. Numbers: $A = 1010 A = 1010$ and $B = 0111 B = 0111$	02	2	3
	1.8	What is a bistable element, and how is it used in digital circuits?	01	1	1
	1.9	Differentiate between a SR latch and a D latch.	01	2	4
	1.10	How does an edge-triggered flip-flop differ from a level-triggered flip-flop?	01	1	2
	1.11	Name the logic component often used self-correcting counters to detect errors?	02	1	2
	1.12	What is the purpose of an assembler directive in assembly language programming?	02	2	4
	1.13	What is the role of the instruction register (IR) in a CPU, and how does it differ from the program counter (PC)?	02	2	4

PART-B

2	a	Explain the Quine-McCluskey method for simplifying Boolean expressions. Use the function: $F(A,B,C,D) = \Sigma(0,1,2,5,8,9,10,14)$.	08	3	3
	b	Explain the $IEEE$ standards used for single & double precision floating point number representation with examples.	08	3	4
3	a	Describe the operation of a D flip-flop and a JK flip-flop implemented as edge-triggered flip-flops. Explain their working with truth tables and timing diagrams.	08	3	4

	b	What is priority encoder? Design a 4:2 priority encoder with necessary diagram.	08	3	3
		OR			
4	a	With a neat diagram explain $4-bit$ parallel adder with accumulator. Provide the truth table for the same.	08	3	5
	b	What is a multiplexer? Write a logic diagram for 8:1 multiplexer using 4 input AND and OR gates.	08	3	4
5	a	Define counter. Design a $mod5$ counter using JK flip flop.	08	3	4
	b	Explain the working of a $3-bit$ shift register with the help of a neat diagram.	08	2	4
		OR			
6	a	Design and explain $3-bit$ binary Up-counter using T flip-flops.	12	4	4
	b	Mention any four applications of shift registers.	04	2	4
7	a	With a neat diagram of connections between the processor and memory, list and explain the basic steps needed to execute the machine instruction $Move\ LOCA, R0$ in terms of transfers between the components of processor, memory and some control commands.	06	3	2
	b	What are addressing modes? Explain any five different types of addressing nodes with examples.	10	2	3
		OR			
8	a	What is a subroutine? Analyze the use of call and return instructions in a subroutine with assembly language program code.	06	3	3
	b	Describe with an example, explain the usage of stack frame in nested subroutine. Also demonstrate how the main program calls subroutine1 and subroutine1 calls subroutine2.	10	4	4
9	a	List the actions needed to execute the instruction $ADD\ (R3), R1$. Write and explain the sequence of control steps for executing the same.	06	2	2
	b	State the importance of cache memory. Describe different types of cache mapping techniques with a neat diagram.	10	2	2
		OR			
10	a	Analyze with an example how the complete execution of the instruction is carried out in the computer system?	06	3	3
	b	With the help of a neat diagram explain the organization of $(1K\ X\ 1)$ memory chip in detail.	10	3	4